

Ground Control Station Concept of Operations (CONOPS)

Project: Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM)

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1. Purpose

The purpose of this Concept of Operations (CONOPS) is to describe the operational use of the Ground Control Station (GCS) as a subsystem of the Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM). The Ground Control Station serves as the primary interface between the operator and the unmanned aircraft system by providing command and control, mission planning, telemetry monitoring, payload observation, and mission data management capabilities.

Unlike the system-level CONOPS, which describes overall mission execution, this document focuses specifically on the operational behavior of the Ground Control Station and its role in supporting flight operations. The GCS enables the operator to configure missions, monitor system status, issue commands, receive aircraft telemetry, observe payload information, and archive mission data throughout all phases of operation.

This document establishes the intended operational concept for the Ground Control Station prototype and serves as a reference for requirements development, architecture definition, verification planning, and subsystem integration.

2. Scope

This CONOPS applies exclusively to the Ground Control Station subsystem of the TOMTOM project. The document describes how the Ground Control Station is expected to operate before, during, and after flight operations while supporting mission planning, command and control, telemetry monitoring, and system management. It identifies the operational functions performed by the GCS, defines the major operational modes, and describes interactions between the operator and supporting hardware and software components.

This document does not describe aircraft flight behavior, autonomous navigation algorithms, payload processing, or flight controller implementation. Those subjects are addressed within the overall TOMTOM system documentation.

3. System Overview

The Ground Control Station is a portable command and control workstation designed to support deployment of the TOMTOM unmanned aircraft system in laboratory, training, and field environments. The system integrates computing hardware, power management, telemetry communications, operator controls, and mission software into a single transportable enclosure.

The Ground Control Station provides the operator with a centralized interface for mission planning, aircraft monitoring, payload observation, and mission execution. During operation, the

GCS continuously exchanges telemetry and command information with the aircraft while displaying mission status and system health information to the operator.

The current prototype utilizes commercial off-the-shelf hardware integrated within a ruggedized equipment case. Internal subsystems include the embedded computing platform, power distribution system, battery power supply, communications equipment, operator interface devices, and mission software environment. These components function together as an integrated subsystem supporting safe and effective operation of the TOMTOM aircraft.

The modular architecture allows hardware components to be upgraded or replaced with minimal impact on the remainder of the system. This approach supports iterative prototype development while providing flexibility for future mission capabilities and subsystem expansion.

4. Operational Concept

The Ground Control Station supports mission execution through five sequential operational phases.

4.1 Mission Preparation

Prior to deployment, the operator transports the Ground Control Station to the operating location and performs a visual inspection of the enclosure, external interfaces, and support equipment. Battery charge status is verified and any required external peripherals are connected. Mission software, mapping data, and aircraft configuration files are reviewed to ensure readiness for operation.

4.2 System Initialization

The Ground Control Station is powered using the integrated battery system and electrical distribution architecture. The embedded computer boots the Linux operating system and launches required mission software including QGroundControl and supporting communication services. Internal subsystem status is verified prior to establishing communication with the aircraft.

4.3 Pre-Flight Verification

After initialization, communication links are established between the Ground Control Station and the aircraft. The operator verifies telemetry reception, GPS status, battery levels, communication signal strength, payload functionality, and overall system health. Mission parameters are reviewed and uploaded as required before authorizing flight operations.

4.4 Mission Execution

During flight operations the Ground Control Station continuously monitors aircraft telemetry, mission progress, payload information, and communication status. The operator issues flight

commands, modifies mission parameters when necessary, and monitors system health throughout the mission. Recorded telemetry and mission data are stored for later analysis.

The Ground Control Station also provides continuous situational awareness through display of aircraft position, flight status, navigation information, battery state, communication quality, and payload imagery. Alerts are presented to the operator whenever abnormal operating conditions are detected.

4.5 Mission Recovery and Shutdown

Upon completion of the mission, the aircraft is recovered and communication links are terminated. Mission data is archived and log files are preserved for analysis. The Ground Control Station performs an orderly software shutdown before removing electrical power from the system. External hardware is disconnected and the equipment is prepared for transportation or storage.

5. Operational Modes

The Ground Control Station operates in several distinct modes throughout mission execution.

Mode	Description
Off	All electrical systems are de-energized.
Initialization	Hardware and software startup sequence.
Standby	System powered and awaiting mission assignment.
Pre-Flight	Verification of communications and subsystem readiness.
Mission	Active command, control, telemetry monitoring, and payload observation.
Recovery	Post-flight data management and aircraft recovery support.
Maintenance	Software updates, hardware inspection, and troubleshooting.
Shutdown	Controlled termination of software and electrical power.

Each operational mode defines a unique system state and establishes the functions available to the operator during that phase of operation.

6. Functional Capabilities

The Ground Control Station provides the following primary operational capabilities:

Mission Planning

Supports mission preparation through configuration of aircraft parameters, mission waypoints, communication settings, and operational planning prior to flight.

Command and Control

Provides the operator with the capability to transmit commands to the aircraft while monitoring command execution and aircraft response throughout mission operations.

Telemetry Monitoring

Displays real-time aircraft telemetry including position, altitude, battery condition, communication quality, flight status, and system health.

Payload Monitoring

Displays payload information including video streams and associated mission data received from the aircraft.

Power Management

Distributes regulated electrical power to Ground Control Station subsystems while monitoring battery condition and overall power system health.

Data Management

Records telemetry, mission logs, operator actions, and system events for future analysis and verification activities.

System Monitoring

Continuously monitors hardware status, communication integrity, software operation, and subsystem health to support reliable mission execution.

7. Interfaces

The Ground Control Station interfaces with multiple internal and external systems throughout mission execution.

Interface	Purpose
Operator	Mission planning, monitoring, and command input.
UAV Flight Controller	Exchange of flight commands and telemetry.
Telemetry Radio	Wireless communication between aircraft and GCS.
Mission Software	Flight planning, monitoring, and vehicle management.
Power Distribution System	Electrical power to internal subsystems.
Payload Video Stream	Reception and display of payload imagery.
External USB Devices	Connection of peripherals and future expansion hardware.

These interfaces collectively enable the Ground Control Station to perform its command, control, monitoring, and mission support functions.

8. Assumptions and Constraints

The Ground Control Station is designed under several operational assumptions. It is assumed that the operator has received appropriate system training, required mission software has been properly configured, communication hardware is operational, and the aircraft has successfully completed all required pre-flight inspections.

The prototype is constrained by the use of commercial off-the-shelf hardware, portable battery operation, Linux-based software, available budget, and the physical dimensions of the Ground Control Station enclosure. These constraints influence system architecture while maintaining sufficient flexibility for future subsystem expansion.

9. Success Criteria

The Ground Control Station is considered to have successfully fulfilled its operational role when it reliably supports mission planning, establishes communication with the aircraft, continuously monitors telemetry and payload information, records mission data, enables operator command and control, and performs an orderly shutdown following mission completion.

Successful operation is further demonstrated through reliable subsystem integration, effective operator interaction, and support of safe mission execution throughout all operational phases.